DOCTORAL PROGRAM IN ENGINEERING SCIENCES AT ITESO

AN UPDATED AND OPEN SOURCED LaTeX TEMPLATE FOR INTERNAL RESEARCH REPORTS OF THE DOCTORAL PROGRAM IN ENGINEERING SCIENCES AT ITESO

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Abstract[1]

An updated and open sourced LaTeX template to write formal internal research reports for the PhD Program in Engineering Sciences at ITESO is presented. It provides a custom document class (`iteso-irr`) compatible with Overleaf, implementing the institutional formatting standard: automated cover page and title page generation, Roman-numeral section numbering, IEEE-style captions and citation format, and BibTeX-based bibliography management with Zotero integration. The template separates content from presentation, allowing authors to focus on technical writing while the class enforces consistent formatting across all reports in the doctoral program.

I. INTRODUCTION

On a first note, this template is based on the formatting standard established by the internal research reports template originally developed by Rayas-Sánchez for the CAECAS research group at ITESO [1], later adopted as the official format for the Doctoral Program in Engineering Sciences. The present LaTeX implementation preserves the same structural and typographic conventions while leveraging automated cross-referencing, bibliography management, and consistent compilation through Overleaf.

In order to make consistent progress on a research line, it is extremely important to make rigorous documentation of every significant step forward. Documenting internal research reports is particularly important within a group of people. Most researchers tend to be quite disorganized

[1]This work was supported in part by SECIHTI (Secretaría de Ciencia, Humanidades, Tecnología e Innovación) through the recognition granted to the PhD Program in Engineering Sciences at ITESO.

with their paper work. Usually, they create many handwriting notes, provisional word-processing files, scripts, and diagrams for each project they develop. After some time (a few months or even a few weeks) it is very difficult to find, understand, and re-use such documented material.

Writing formal internal reports is also important to show evidence on research progress to other organizations. In Mexico, for instance, SECIHTI requires formal internal reports to most research groups that are granted significant research funding. In addition, by having an official internal report we can more easily deal with intellectual property and copyright issues.

Internal reports do not have to be long. A report might consist of just 5 or 6 pages, including cover page and figures. What is most important of an internal research report is its accuracy in terms of contents and consistency in terms of format. Making research is similar to building a house: it is realized by assembling many simple building blocks. Each building block must be robust and well built. Similarly, every internal report must have been extensively reviewed and revised, so that it becomes a useful piece of knowledge for further research work.

By having a simple and well-defined template for writing research reports, authors can focus their attention on the technical issues (logical argumentation, clarity, precision, consistency, etc.) and writing aspects (grammar, spelling, punctuation, etc.), since most of the formatting is realized semi-automatically. A suitable template should facilitate allocating figures and formatting equations and references.

This report describes the format and style that must be used for internal research reports in the Doctoral Program in Engineering Sciences at ITESO, particularly to document progress in the Seminarios IDI (Seminars on Research, Development, and Innovation) for this PhD program.

The internal research reports for the PhD Program in Engineering Sciences at ITESO must be written in English, to facilitate sharing information and scientific publishing. It should also be noticed that text in other languages different to English must be typed either between quotation marks, or using italics (e.g., see footnote on previous page).

II. USING LaTeX STYLES

This template uses a custom document class, `iteso-irr`, which encapsulates all formatting rules. Each structural element of the document is handled by a specific LaTeX command or environment. Normal text is typeset using the default body font and paragraph settings, while the Roman-numbered section titles are produced by the `\section{}` command. Subsections use the `\subsection{}` command and are labeled with capital letters.

When a portion of text is copied from another file, it can be pasted directly. LaTeX will automatically apply the formatting defined in this template. In other words, authors write plain text with standard LaTeX markup in `main.tex`, and the class file enforces the institutional format. No

manual formatting of fonts, margins, headers, or caption styles is needed.

III. DOCUMENT STRUCTURE

The present template provides all the necessary commands for structuring the document. The cover page is produced by `\makecoverpage`, and the first page (with keywords, abstract, and funding acknowledgment) is produced by `\makefirstpage`.

Report metadata is declared once in the preamble of `main.tex` through commands such as `\ReportTitle`, `\ReportAuthors`, `\ReportFileName`, `\ReportDate`, and `\CopyrightYear`. All occurrences throughout the document (cover page, title page, headers, footers) are updated automatically upon recompilation.

IV. INITIAL PAGES

A. The Cover Page

The cover page contains the report title, author name(s), the filename identifier, the date and location, the institutional affiliation, the ITESO logo, and the copyright notice. The filename follows the convention `PhDEngScITESO-YY-NN-R`, where `YY` corresponds to the last two digits of the year of publication and `NN` corresponds to a consecutive number assigned by the Chair of the PhD program.

To insert the ITESO logo, place the image file (e.g., `iteso-logo.png`) in the style/ project directory and uncomment the corresponding `\includegraphics` line in the cover page definition inside `iteso-phdengsci.def`.

B. The First Page

The first page automatically displays the title, authors, institutional contact information, keywords, and abstract. When the reported research work has been supported financially, the corresponding credit is given using a footnote attached to the abstract header. The funding text is passed as the third argument to `\makefirstpage`.

Keywords are one-word or two-word expressions representing the main topics addressed in the report. All keywords should be in lower-case, except proper names and acronyms.

V. INSERTING EQUATIONS AND MATHEMATICAL SYMBOLS

All mathematical symbols must be italic, including Greek letters. Vectors and matrices must be bold italic, and scalars must use regular italic font (not bold). This is an example of a linear system:

$$\boldsymbol{A}\boldsymbol{y} + b\boldsymbol{z} = \boldsymbol{c} \tag{1}$$

where $\boldsymbol{A} \in \Re^{n\times m}$, $\boldsymbol{y} \in \Re^{m}$, $\boldsymbol{z}$ and $\boldsymbol{c} \in \Re^{n}$, $b \in \Re$. Notice that the mathematical expressions in this paragraph are inserted as inline math and not as plain text. This ensures consistent font style with the displayed equations.

An example from machine learning is the regularized empirical risk minimization objective:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \frac{1}{N}\sum_{i=1}^{N} \mathscr{L}\Big(f(\boldsymbol{x}^{(i)};\boldsymbol{\theta}), y^{(i)}\Big) + \lambda \|\boldsymbol{\theta}\|_2^2 \tag{2}$$

where $\boldsymbol{\theta} \in \Re^{d}$ contains the model parameters, $\{(\boldsymbol{x}^{(i)}, y^{(i)})\}_{i=1}^{N}$ is the training set, $\mathscr{L}$ denotes the loss function, $f(\cdot;\boldsymbol{\theta})$ is the model, and $\lambda \geq 0$ controls the strength of L_2 regularization.

Numbered equations use the `equation` environment, as in (1) and (2), so that they are centered and the equation number is aligned to the right. To refer to an equation, use `\eqref{}`. For example: "from (1) we obtain…", instead of "from equation 1 we obtain…".

Simple mathematical expressions can be typed directly as part of the text body using inline math mode, as long as they do not need to be referenced later. An example: $\beta = \alpha/(1-\alpha)$.

TABLE I
SUMMARY OF REFERENCE TYPES AND THEIR FORMATS

Reference Type	**Example Citation**
Journal papers	[2]
Books	[3]
Book chapters	[4]
Conference papers (ML)	[5]
Conference papers (Eng.)	[6]
arXiv preprints	[7]
Technical reports	[8]
Datasets	[9]
Software repositories	[10]
Open source software	[11]
Internal research reports	[1]

TABLE II
EXAMPLE OF AN EXPERIMENTAL RESULTS TABLE

Method	**Accuracy (%)**	**F1 Score**	**Time (s)**
Baseline SVM	82.3 ± 0.5	0.791 ± 0.008	12.4
Random Forest	85.1 ± 0.4	0.823 ± 0.006	8.7
Neural Network	$\mathbf{89.7 \pm 0.3}$	$\mathbf{0.881 \pm 0.004}$	45.2

VI. FIGURES AND TABLES

A. Tables

In LaTeX, figures and tables are handled as floating environments that are automatically placed by the typesetting engine within the main document. Placement can be influenced through optional specifiers (`h`, `t`, `b`, `p`) passed to the environment.

Tables are always shown at the beginning of the file containing figures and tables, and are numbered using Roman numerals. Their captions are placed above the table and use small capitals. Table I illustrates the standard format for citation examples. For the case of having two or more tables to shown, it might be needed a `\clearpage` command to start the section or subsection in a new page. And, When presenting experimental results, it is conventional to bold-face the best result in each column and to report uncertainty as mean $\pm$ standard deviation. Table II shows the recommended format. Also, The text included within a table must be horizontally and vertically centered. Regular rows do not use visible horizontal lines beyond the header and footer rules. Columns should never use visible vertical lines.

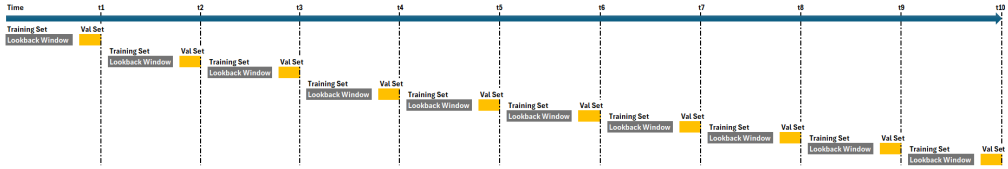

FIG. 1: EXAMPLE FIGURE SHOWING TYPICAL FORMAT FOR PLOTS

B. Figures

Each figure must have a descriptive caption (as self-contained as possible), so that the reader can understand the work by examining the figures and reading their captions. Figures are inserted using the `figure` environment and the `\includegraphics` command, as it can be appreciated in with the Fig 1, which is a wide horizontally oriented plot.

When inserting figures with multiple images, use the `subcaption` package with the `subfigure` environment, providing identifiers (a), (b), etc., for each sub-image.

C. Inserting Figures from Matlab

Figures created in Matlab should be exported in vector format (PDF or EPS) for best results in LaTeX. The recommended Matlab commands are:

```
print('-dpdf', 'figure_name.pdf')
% or
exportgraphics(gcf, 'figure_name.pdf', ...
    'ContentType', 'vector')
```

Listing 1: Exporting a matplotlib figure to PDF.

All plots should be distinguishable in both color and black-and-white printing, by using not only different colors but also different line styles and markers.

D. Inserting Figures from Python and R

Figures created with Python's `matplotlib` or `seaborn` libraries should be saved in PDF format for vector quality:

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()
ax.plot(x, y, label='Training loss')
ax.set_xlabel('Epoch')
ax.set_ylabel('Loss')
ax.legend()
```

```
fig.savefig('figure_name.pdf', bbox_inches='tight')
```

Listing 2: Exporting a matplotlib figure to PDF.

For R, the equivalent export using `ggplot2` is:

```
library(ggplot2)
p <- ggplot(data, aes(x = epoch, y = loss)) +
     geom_line() +
     labs(x = "Epoch", y = "Loss")
ggsave("figure_name.pdf", plot = p,
       width = 6, height = 4)
```

Listing 3: Exporting a ggplot2 figure to PDF.

Figures generated inside Jupyter notebooks can be saved to PDF programmatically by calling `fig.savefig()` in the cell that produces the plot. The saved PDF file is then included in the LaTeX project via `\includegraphics` as with any other figure.

E. Inserting Figures from Other Tools

Figures created with drawing tools such as Visio, Inkscape, or `draw.io` should be exported in vector format (PDF or SVG) before inclusion. PDF is the preferred format for `pdflatex` compilation. If only SVG is available, convert it to PDF using Inkscape on the command line:

```
inkscape --export-type=pdf figure_name.svg
```

Bitmap formats (PNG, JPEG) should be used only when no vector alternative exists, such as photographs or screenshots. Bitmap figures consume more memory and lose resolution when scaled.

F. Creating Figures with TikZ and PGFPlots

For figures that must match the document font exactly (architecture diagrams, flowcharts, training curves), the `tikz` and `pgfplots` packages allow figures to be defined directly in LaTeX source code. This approach produces figures whose labels, symbols, and fonts are rendered by the same typesetting engine as the body text. The `tikz` and `pgfplots` packages are not loaded by default in this template to avoid increasing compilation time for reports that do not use them. To enable them, add the following lines in the preamble of `main.tex`:

```
  \usepackage{tikz}
  \usepackage{pgfplots}
  \pgfplotsset{compat=1.18}
```

VII. ALGORITHMS AND CODE LISTINGS

A. Algorithm Pseudocode

The `algorithm2e` package (loaded by the document class) provides a structured environment for presenting algorithms. Algorithms are numbered independently of equations and figures, and can be referenced with `\label{}` and `\ref{}`.

Algorithm 1 shows an example of stochastic gradient descent in the format expected for internal research reports.

Algorithm 1: Mini-Batch Stochastic Gradient Descent

Input: Training set $\mathscr{D} = \{(\boldsymbol{x}^{(i)}, y^{(i)})\}_{i=1}^{N}$, learning rate η, number of epochs T
Output: Trained parameters $\boldsymbol{\theta}$

Initialize $\boldsymbol{\theta}$ randomly;
for $t \leftarrow 1$ **to** T **do**
  Shuffle $\mathscr{D}$;
  for *each mini-batch* $\mathscr{B} \subset \mathscr{D}$ **do**
    Compute gradient $\boldsymbol{g} \leftarrow \frac{1}{|\mathscr{B}|} \sum_{(\boldsymbol{x},y)\in\mathscr{B}} \nabla_{\boldsymbol{\theta}} \mathscr{L}(f(\boldsymbol{x};\boldsymbol{\theta}), y)$;
    Update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\, \boldsymbol{g}$;
  end
end
return $\boldsymbol{\theta}$;

The key elements demonstrated above are: `\KwIn` and `\KwOut` for specifying inputs and outputs, `\For` for loop constructs, mathematical notation within the pseudocode body, and `\caption` plus `\label` for referencing. Conditional constructs use `\If`, `\ElseIf`, and `\Else`. The `algorithm2e` documentation provides the full set of available constructs.

B. Source Code Listings

The `listings` package (loaded by the document class) provides formatted source code blocks with syntax highlighting and line numbers. Listings are appropriate when the exact implementation matters, such as a data preprocessing pipeline or a model architecture definition. Listing 4 shows an example.

```
import torch
import torch.nn as nn

class SimpleNet(nn.Module):
    def __init__(self, input_dim, hidden_dim, output_dim):
        super().__init__()
```

```
        self.fc1 = nn.Linear(input_dim, hidden_dim)
        self.relu = nn.ReLU()
        self.fc2 = nn.Linear(hidden_dim, output_dim)

    def forward(self, x):
        x = self.relu(self.fc1(x))
        return self.fc2(x)
```

Listing 4: A simple feedforward neural network in PyTorch.

The `lstset` defaults (defined in `iteso-phdengsci.def`) provide a single-frame style with line numbers. These defaults can be overridden per listing through optional parameters in the `\begin{lstlisting}` environment.

For short inline code references within the text body, use `\texttt{}` for identifiers (e.g., `nn.Module`) or `\lstinline` for syntax-highlighted fragments (e.g., `def forward(self, x):`).

VIII. MAKING REFERENCES

It is very important to make proper reference to relevant work of other authors, as well as to our own previous related work. The format used in the PhD Program in Engineering Sciences at ITESO is the IEEE citation reference format [12].

The following are examples of the format for different types of references:

a) Journal papers: [13], [2].
b) Books: [3], [14].
c) Book chapters: [4], [15].
d) Conference papers (engineering): [6], [16].
e) Conference papers (ML venues): [5], [17].
f) arXiv preprints: [7], [18].
g) Technical reports (industry labs): [8], [19].
h) Datasets: [9], [20].
i) Software repositories: [10], [21].
j) Open source software and languages: [11], [22], [23], [24].
k) Handbooks or manuals: [25], [26].
l) Standards: [27].
m) Patents: [28], [29].
n) Online sources: [12], [30].

o) Thesis: [31], [32].

p) Internal research reports: [1], [33].

A. Cross References

In LaTeX, cross references are handled automatically via the `\label{}` and `\ref{}` (or `\eqref{}`, `\cite{}`) mechanisms. All references are updated upon recompilation. There is no need for manual renumbering.

B. References to Software Tools

References to specific commercial or public-domain software tools should be inserted as footnotes and not as bibliographic references. For example: MATLAB[2], PyTorch[3], scikit-learn[4], and Docker[5]. It is important to indicate the specific version of the software used, since it might have an impact on the reported technical results.

C. References to Websites

References to websites of organizations or individuals may be inserted as footnotes. Several examples are: MOSIS Educational Program[6], a research group website[7], and a government website[8]. Since information on websites is volatile, it is important to include the date when the information was last accessed.

Alternatively, web references with sufficient identification data (author, title, year, URL) may be included as standard bibliography entries using the `@misc` or `@online` entry type in BibTeX. Zotero captures web page metadata automatically and exports it in the correct format through Better BibTeX.

D. Using a Bibliographic Database

Bibliography management is handled by BibTeX. All references are stored in `references.bib` using standard entry types (`@article`, `@book`, `@inproceedings`, `@phdthesis`, etc.). The IEEE citation style is applied automatically via the `IEEEtran` bibliography style file. Adding a new reference requires only creating a new entry in the `.bib` file; numbering, formatting, and sorting are handled automatically upon recompilation.

[2]MATLAB, Version R2024a, The MathWorks, Inc., Natick, MA, USA, 2024.

[3]PyTorch, Version 2.2.0, Meta Platforms, Inc., 2024. `https://pytorch.org/`.

[4]scikit-learn, Version 1.4.0, 2024. `https://scikit-learn.org/`.

[5]Docker Engine, Version 25.0, Docker, Inc., Palo Alto, CA, USA, 2024. `https://www.docker.com/`.

[6]The MOSIS Service, MOSIS Educational Program. Mar. 21, 2007, `http://www.mosis.com/products/mep`.

[7]ESIME-Zacatenco-IPN, EMC Group. Apr. 3, 2015, `http://www.sepi.esimez.ipn.mx/lab-emc/`.

[8]Banco Nacional de Comercio Exterior, Información Sectorial. Oct. 5, 2002, `http://www.bancomext.com`. (no longer available).

The recommended workflow is to manage references in Zotero and export them to BibTeX format using the Better BibTeX plugin, which can be configured to keep a `.bib` file continuously synchronized with a Zotero collection, so that any reference added or edited in Zotero is immediately available in the LaTeX project. In Overleaf, the synchronized `.bib` file can be linked through GitHub integration or uploaded directly. This eliminates manual transcription of references and ensures consistent formatting across all entries.

To set up this workflow:

a) Install the Better BibTeX plugin in Zotero.

b) Right-click a Zotero collection and select "Export Collection", choosing "Better BibTeX" as the format and enabling "Keep updated".

c) Save the exported file as `references.bib` in the project directory.

d) In Overleaf, upload the file or synchronize it via a linked GitHub repository.

IX. APPENDIX OF FILES USED

There are usually many files associated with the creation of an internal research report. These files can be Python scripts, Jupyter notebooks, Matlab files, R scripts, configuration files, trained model weights, datasets, SPICE circuit files, Visio files, Dockerfiles, or shell scripts for experiment orchestration. These filenames should be documented as part of the internal research report.

The list of extra filenames should be inserted here. This appendix should not include the basic files (`PhDEngScITESO-26-00-R.tex`, `references.bib`, and the compiled PDF). It should only include additional files.

A. Example

- `train.py` – Training script for the neural network.
- `evaluate.py` – Evaluation and metrics computation.
- `analysis.ipynb` – Jupyter notebook with exploratory data analysis and result visualization.
- `config.yaml` – Hyperparameter configuration file.
- `model_best.pt` – Best model checkpoint (PyTorch).
- `preprocess.R` – R script for data preprocessing.
- `simulation_results.m` – Matlab script for Fig. 1.
- `circuit_schematic.vsdx` – Visio drawing for Fig. 2.
- `Dockerfile` – Container definition for reproducibility.
- `run_experiments.sh` – Shell script to orchestrate training runs.

X. REPRODUCIBILITY

When the reported work involves computational experiments, the internal research report should include sufficient information to allow independent reproduction of the results. A reproducibility statement serves this purpose by documenting the complete experimental setup in a structured format. The following categories should be addressed as applicable.

A. Software Environment

State the programming language and its exact version (e.g., Python 3.12.2). List all libraries and frameworks that directly affect the results, with their version numbers (e.g., PyTorch 2.2.0, scikit-learn 1.4.0, NumPy 1.26.4). If a package manager lock file or a `requirements.txt` is provided, reference it in the Appendix of Files Used (Section IX). For containerized environments, specify the base image and include the `Dockerfile`.

B. Hardware and Compute Resources

Describe the hardware on which experiments were executed: GPU model and memory (e.g., NVIDIA A100 80 GB), CPU architecture and core count, and total system memory. For cloud-based experiments, state the instance type and provider (e.g., AWS `p4d.24xlarge`). Report the total wall-clock time for the main experiments so that readers can estimate the computational cost of reproduction.

C. Stochastic Control

Specify the random seed strategy used across all stochastic components: weight initialization, data shuffling, dropout masks, and any sampling-based procedures. State the exact seed value (e.g., `seed=42`) and whether deterministic mode was enabled in the framework (e.g., `torch.use_deterministic_algorithms(True)`). If full determinism is not achievable (as with certain GPU operations), document the known sources of non-determinism and the observed variance across repeated runs.

D. Data

Identify the dataset by name, version, and source URL. Describe any preprocessing, filtering, or augmentation steps applied. State the exact train/validation/test split ratios and the method used to generate them (e.g., stratified random split with a fixed seed). If the dataset is not publicly available, describe the access procedure and any restrictions. For generated or synthetic data, include the generation script in the Appendix of Files Used.

E. Statement Template

The following block can be use as a template to later be adapted for each report:
All experiments were conducted using Python 3.12 and PyTorch 2.2.0 on a single NVIDIA A100 GPU (80 GB). Random seeds were fixed to 42 for weight initialization, data shuffling, and dropout. Deterministic algorithms were enabled via the torch instruction for deterministic algorithms. The training dataset [name, version] is publicly available at [URL]. Preprocessing consisted of [brief description]. The train/validation/test split was [ratio], generated by stratified random sampling with the same seed. Total training time was approximately [X] GPU-hours across [N] runs. The complete source code, configuration files, and trained model checkpoints are listed in the Appendix of Files Used (Section IX).

XI. CONCLUSIONS

Conclusions should summarize the main ideas and general results described in the internal research report. Conclusions should be brief, and should be consistent with the abstract and the introduction. It is also a good idea to report in the conclusions the research problems that should be investigated in the near future, i.e., problems or questions that were detected during the documented research work, whose solution or answer is still not clear, that require more investigation, and that seem to be relevant for future contributions.

ACKNOWLEDGEMENT

Acknowledgement should be included if the work has been technically supported by an individual or an external organization. For instance, when the work reported has been realized using a software tool with a special license (donated license, academic version, etc.), the corresponding company should be recognized.

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